Southern Nuclear

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July 3, 2024

Docket Nos.: 50-321
50-366

NL-24-0237

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D. C. 20555-0001

Edwin I. Hatch Nuclear Plant – Unit 1 & 2
Licensee Event Report 2024-002-01
Incorrectly Installed Temporary Modification Results in Multiple Conditions Prohibited by Plant Technical Specifications

Ladies and Gentlemen:

In accordance with the requirements of 10 CFR 50.73 (a)(2)(i)(B), Southern Nuclear Operating Company hereby submits the enclosed Licensee Event Report.

This letter contains no NRC commitments. If you have any questions, please contact the Hatch Licensing Manager, Jimmy Collins, at 912.453.2342.

Respectfully submitted,

Matt Busch
Vice President – Hatch

MSB/CJC

Enclosure: LER 2024-002-01

Cc: Regional Administrator, Region II
NRR Project Manager – Hatch
Senior Resident Inspector – Hatch
RTYPE: CHA02.004

Edwin I. Hatch Nuclear Plant Unit 1 & 2
Licensee Event Report 2024-002-01

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Enclosure

LER 2024-002-01

NRC FORM 366 (04-02-2024)	U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104	EXPIRES: 04/30/2027

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name		2. Docket Number	3. Page
Edwin I. Hatch Nuclear Plant Unit 2	☒ 050 / ☐ 052	00366	1 OF 4

4. Title

Incorrectly Installed Temporary Modification Results in Multiple Conditions Prohibited by Plant Technical Specifications

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name		Docket Number
02	27	2024	2024	- 002 -	01	07	03	2024	EI Hatch Unit 1	☒ 050	00321
									Facility Name	☐ 052	Docket Number

9. Operating Mode	10. Power Level
MODE 1	100%

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☑ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Jimmy Collins - Licensing Manager	9124532342

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
A	VI	CDU	T265	Y					

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☑ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

At 1301 EST on 2/27/2024 while Unit 2 was at 100% power in MODE 1 and Unit 1 was at 0% power in MODE 4, it was discovered that a temporary Plant Service Water (PSW) modification was incorrectly installed on the Bravo condensing unit (CDU) of the Main Control Room Air Conditioning (MCRAC) subsystem. This temporary modification was installed on 1/30/2024 and then removed on 2/28/2024.

For the time period that the incorrectly installed temporary PSW modification was in place, a past operability review (POR) was performed. The POR confirmed that the temporary PSW modification was installed incorrectly, rendering the Bravo MCRAC subsystem inoperable for a period of 29 days. Additionally, it was determined that the Alpha MCRAC subsystem was inoperable due to low refrigerant between 1/25/2024 and 2/8/2024. Due to these inoperabilities, several conditions prohibited by Technical Specifications (TS) were experienced as discussed below. With the removal of the temporary PSW modification on 2/28/2024 the operability of the Bravo MCRAC subsystem was restored. The Alpha MCRAC subsystem was repaired by adding refrigerant on 2/8/2024.

NARRATIVE

EVENT DESCRIPTION

At 1301 EST on 2/27/2024 while Unit 2 was at 100% power in MODE 1 and Unit 1 was at 0% power in MODE 4, it was discovered that a temporary Plant Service Water (PSW) (EIIS Code: BI) modification was incorrectly installed on the Bravo condensing unit (CDU) of the Main Control Room Air Conditioning (MCRAC) subsystem (EIIS Code: VI). This temporary modification was installed on 1/30/2024 @ 1020 and then restored on 2/28/2024 @ 1441.

For the time period that the incorrectly installed temporary PSW modification was in place, a past operability review (POR) was performed. The POR confirmed that the temporary PSW modification was installed incorrectly, rendering the Bravo MCRAC subsystem inoperable for a period of 29 days, between 1/30/2024 @ 1020 and 2/28/2024 @ 1441. Additionally, it was determined that the Alpha MCRAC subsystem was inoperable due to low refrigerant between 1/25/2024 @ 0416 - 2/8/2024 @ 0410. Due to these inoperabilities, twelve conditions prohibited by Technical Specifications (TS) were experienced as discussed below.

For reference, Unit 1 and Unit 2 share MCRAC and Main Control Room Environmental Control (MCREC) subsystems and have a common refueling floor. Unit 2 was in MODE 1 throughout the duration of these events. Unit 1 was in MODE 1 until 2/4/2024 @ 0008 when it entered MODE 3. Unit 1 was cooled down to MODE 4 on 2/4/2024 @ 0653 and then entered MODE 5 on 2/5/2024 @ 1211. Unit 1 transitioned back to MODE 4 on 2/23/2024 @ 1850.

Prohibited Condition 1 & 2: Unit 1 & Unit 2 were not brought to MODE 3 in 12 hours as required by TS 3.7.4.E

The design of the MCREC system requires one operable MCRAC subsystem per MCREC subsystem. With the Bravo MCRAC subsystem inoperable, the other two subsystems (Alpha and Charlie) have to be in-service to meet the requirements of the MCREC TS condition 3.7.4. The POR revealed that this condition was not continuously met. From 2/1/2024 @ 0824 until 2/1/2024 @ 2147 all three MCRAC units were inoperable. Alpha and Bravo were inoperable for reasons discussed above and Charlie was inoperable due to a cooling water tagout. This condition, which resulted in both MCREC trains being rendered inoperable, existed in excess of the Unit 1 & Unit 2 TS condition 3.7.4.E (two MCREC subsystems inoperable) completion time of 12 hours to be in MODE 3.

Prohibited Conditions 3 & 4: Unit 2 was not brought to MODE 3 in 12 hours as required by TSs 3.7.4.C and 3.7.5.D

Once the Bravo MCRAC unit was rendered inoperable due to the PSW modification, concurrent with the Alpha MCRAC inoperability, one of the MCREC subsystems was rendered inoperable. The required actions for Unit 2 TS conditions 3.7.4.A (one MCREC inoperable) and 3.7.5.B (two MCRAC subsystems inoperable) each have a 7 day completion time. These completion times began on 1/30/2024 @ 1020 and expired on 2/6/2024 @ 1020 requiring entry into TS conditions 3.7.4.C and 3.7.5.D, each with a 12 hour completion time to be in MODE 3. These completion times expired on 2/6/2024 @ 2220, but Unit 2 was not brought into MODE 3.

Prohibited Condition 5: Unit 2 was not brought to MODE 3 in 12 hours as required by TS 3.7.4.E

From 2/6/2024 @ 2350 until 2/8/2024 @ 0410, all three MCRAC units were inoperable (Note: Unit 1 was not in a MODE of applicability specified in TS 3.7.4.E during this time). Alpha and Bravo were inoperable for reasons discussed above and Charlie was inoperable due to an electrical bus tagout. This condition, which resulted in both MCREC trains being rendered inoperable, existed in excess of the Unit 2 TS condition 3.7.4.E (two MCREC subsystems inoperable) completion time of 12 hours to be in MODE 3.

NARRATIVE

EVENT DESCRIPTION (Continued)

Prohibited Conditions 6, 7, 8, 9, 10 & 11: Fuel moves continued in conflict with Unit 1 and Unit 2 TSs 3.7.4.D, 3.7.4.F and 3.7.5.F

Additional conditions prohibited by TS existed related to fuel movements that were in progress between 2/7/2024 @ 2342 and 2/9/2024 @ 2200 when Unit 1 and Unit 2 TS 3.7.4.D, 3.7.4.F and 3.7.5.F required fuel movements to stop immediately. Unit 1 and Unit 2 TS conditions 3.7.4.A and 3.7.5.B had expired on 2/6/2024 @ 1020 and were not exited until 2/10/2024 @ 1415. Also, Unit 1 and Unit 2 TS condition 3.7.4.F was applicable due to having two MCREC units inoperable between 2/6/2024 @ 2350 and 2/8/2024 @ 0410.

Prohibited Condition 12: Unit 2 was not brought to MODE 3 in 12 hours as required by TS 3.7.5.D

The overlapping time of the inoperabilities of the Alpha and Bravo MCRAC subsystems exceeded the allowable 30 day completion time of the Unit 2 TS condition 3.7.5.A (one MCRAC subsystem inoperable) plus the TS condition 3.7.5.D completion time of 12 hours to be in MODE 3. TS condition 3.7.5.A began on 1/25/2024 @ 0416 and expired on 2/24/2024 @ 0416 requiring entry into TS condition 3.7.5.D which expired on 2/24/2024 @ 1616. However, Unit 2 was not brought to MODE 3 in 12 hours as required.

FAILED COMPONENTS INFORMATION

Master Parts List Number: 1Z41B008A & B
Manufacturer: Trane
Model Number: QB-J299-1
Type: Control Room Condensing Unit

EVENT CAUSE ANALYSIS

The direct cause of the incorrect installation of the temporary PSW modification on the Bravo MCRAC, which rendered it inoperable, was a failure of non-licensed, contract maintenance personnel to use Human Performance (HU) tools to ensure the temporary PSW modification was properly installed per the procedure. Contributing to this, the procedure did not contain adequate steps or guidance for certain aspects of the installation of the temporary PSW modification. The cause of the Alpha MCRAC inoperability was low refrigerant.

SAFETY ASSESSMENT

Due to the issues discussed above, these events are reportable per 10 CFR 50.73 (a)(2)(i)(B) as conditions prohibited by TS. There were no actual safety consequences as a result of this event. At any time while the incorrectly installed temporary modification was installed on the Bravo MCRAC subsystem, either Alpha or Charlie MCRAC subsystems were running or capable of being manually aligned and started to ensure the habitability of the control room was maintained. Additionally, the Bravo MCRAC system (and therefore at least one MCREC subsystem) was fully functional other than its seismic capability. Given Hatch's seismic zone location, the risk of this scenario is minor. Also, during winter months, the MCRAC heat sink (plant service water) is at a lower temperature, and therefore only one MCRAC subsystem is typically needed to maintain control room cooling.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Edwin I. Hatch Nuclear Plant Unit 2	■ 050 / ☐ 052	00366	2024	- 002	- 01

NARRATIVE

CORRECTIVE ACTIONS

With the removal of the temporary PSW modification on 2/28/24 the operability of the Bravo MCRAC subsystem was restored. Although the temporary PSW modification was installed on the other two MCRAC subsystems, a walkdown of these subsystems was performed and no issues were identified. Additional completed corrective actions include: (1) this scenario was added to the Craft On-Boarding Ranking Assessment (COBRA) proficiency assessment, and (2) an SNC Cognizant Person Hold Point / Sign Off was added in the job plan for this work evolution to validate proper implementation. Lastly, there is an open action to update the applicable procedure with better detailed instructions for installing the temporary PSW modification. The Alpha MCRAC inoperability was corrected on 2/8/2024 by checking the unit for refrigerant leaks and then restoring proper refrigerant level.

PREVIOUS SIMILAR EVENTS

None.